

Yuhang Lu

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Education

The University of Hong Kong

Ph.D. in School of Computing and Data Science

Hong Kong, China

Oct. 2025 – Sep. 2029 (expected)

Beijing University of Posts and Telecommunications / Queen Mary University of London

Beijing, China / London, UK

Dual-degree B.Eng. in Internet of Things Engineering; GPA: 87.9/100; First Class Honours

Sep. 2018 – Jun. 2022

Research Interests

- Physical AI, especially humanoid robotics and autonomous driving.
- Humanoid robot foundation models with transferability, compositionality, and contact awareness.
- 3D understanding under distribution shift, limited supervision, and long-tail scenarios.

Work Experience

Booster Robotics

Research Intern

Apr. 2025 – Present

Beijing, China

- Worked on humanoid soccer control for dynamic ball interaction, from incoming-ball perception to goal-directed kicking.
- Trained and deploy policy that can coordinate approach, balance, contact timing, and target-point kick accuracy under varying ball trajectories on Booster K1.
- Investigating unified policy learning for multiple soccer skills, such as kicking and ball stopping, through shared perception and skill-conditioned control.

Huawei Technologies Co., Ltd.

Research Intern, supervised by Jiaxin Peng

Jan. 2026 – Apr. 2026

Shenzhen, China

- Contributed to the World Engine project, focusing on behavior world model inference for scenario rollout and agent behavior prediction.
- Designed inference workflows for behavior-conditioned simulation, developing multiple inference mode, performed trajectory feasibility checks, and automated data generation.
- Conducted preliminary research for Huawei ADS and helped design a video-generation-based simulator for scalable autonomous-driving scenario synthesis and evaluation.

OpenDriveLab, Shanghai AI Lab

Research Intern, supervised by Li Chen

Jun. 2025 – Sep. 2025

Shanghai, China

- Researched generative world models for autonomous driving, focusing on controllable scenario synthesis for planning-oriented data generation.
- Generated diverse driving scenes with controllable traffic participants, road context, and interaction patterns to enrich long-tail training data.
- Used generated scenarios for planner post-training and analyzed robustness and generalization under rare and safety-critical cases.

ShanghaiTech University

Research Assistant, supervised by Yuexin Ma

Aug. 2022 – Apr. 2025

Shanghai, China

- Led research on 3D perception and reasoning for autonomous driving, spanning zero-shot point cloud segmentation, camera-based occupancy prediction, driving-knowledge evaluation, and end-to-end planning.
- Designed label- and memory-efficient methods, including multimodal fusion for zero-shot point cloud segmentation and octree-based occupancy querying for camera-only scene understanding.

- Built driving-knowledge benchmarks and human-like planning models, coordinating implementation, experiments, and manuscript writing across multi-author projects.

Publications

1. Tianyu Li*, Li Chen*, Caojun Wang*, Haochen Liu*, Kashyap Chitta, **Yuhang Lu**, Naisheng Ye, Yufei Wang, Jiaxin Peng, Jin Pan, Zhaoyu Su, Andrei Bursuc, Shengbo Eben Li, Andreas Geiger, Honglin Bian, Hongyang Li, Peng Su. **WorldEngine: Towards the Era of Post-Training for Physical AI**. *arXiv*, 2026.
2. Enhui Ma*, Jiahuan Zhang*, Guantian Zheng, Tao Tang, Shengbo Eben Li, **Yuhang Lu**, Xia Zhou, Xueyang Zhang, Yifei Zhan, Kun Zhan, Zhihui Hao, XianPeng Lang, and Kaicheng Yu. **DriveCombo: Benchmarking Compositional Traffic Rule Reasoning in Autonomous Driving**. *CVPR*, 2026.
3. **Yuhang Lu**, Jiangnan Shao, Xinge Zhu, and Yuexin Ma. **ReAL-AD: Towards Human-Like Reasoning in End-to-End Autonomous Driving**. *ICCV*, 2025.
4. Ziyang Yan, Wenzhen Dong, Yihua Shao, **Yuhang Lu**, Haiyang Liu, Jingwen Liu, Haozhe Wang, Zhe Wang, Yan Wang, Fabio Remondino, and Yuexin Ma. **RenderWorld: World Model with Self-Supervised 3D Label**. *ICRA*, 2025.
5. **Yuhang Lu***, Yichen Yao*, Jiadong Tu*, Jiangnan Shao*, Yuexin Ma, and Xinge Zhu. **Can LVLMS Obtain a Driver's License? A Benchmark Towards Reliable AGI for Autonomous Driving**. *AAAI*, 2025.
6. **Yuhang Lu**, Xinge Zhu, Tai Wang, and Yuexin Ma. **OctreeOcc: Efficient and Multi-Granularity Occupancy Prediction Using Octree Queries**. *NeurIPS*, 2024.
7. **Yuhang Lu***, Qi Jiang*, Runnan Chen, Yuenan Hou, Xinge Zhu, and Yuexin Ma. **See More and Know More: Zero-shot Point Cloud Segmentation via Multi-modal Visual Data**. *ICCV*, 2023.

* Equal contribution

Miscellaneous

Languages: Chinese (native), English (IELTS 7)

Technical Toolkit: Python, Git, PyTorch, Linux/Shell, L^AT_EX

Academic Services: Reviewer for CVPR, ICCV, ECCV, NeurIPS, AAAI, ICRA, etc